

Day – 28

Task – Advanced classification and tuning

Day - Monday

Date – 27th April 2026

Time – 7:00 AM – 10:00 AM

Dataset – Wine_Quality_Dataset.

Source – The dataset was downloaded from Kaggle.

Why this dataset - I chose this dataset because it is clean, has only numerical data, and is easy to use for classification and improving models.

- Downloaded the dataset and imported into **Jupyter lite** for cleaning.
- The dataset has **1143 rows** and **13 columns**.
- Imported necessary libraries and dataset using **function**.
- Performed basic operations to get data insights using function.
- Dropped column.
- Selected features to train models.
- Performed decision tree and printed accuracy.
- Performed decision tree tuning
- Performed random forest model.
- Got some errors while performing random forest tuning.
- Saved clean dataset.